Neural Collapse in Terminal Training:

A Small-Scale Reproduction on MNIST with a Mini-ResNet

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Abstract

This report studies the research paper:*Prevalence of Neural Collapse during the Terminal Phase of Deep Learning Training* by Papyan, Han, and Donoho. The paper claims that after a classifier reaches zero training error, continued cross-entropy training drives the final-layer representation toward a simple geometry coined *neural collapse*. This appears through four linked effects that show up in TPT. Observed consistently, on a variety of networks, architectures and datasets.

I reproduced the main qualitative phenomenon on MNIST using a compact Mini-ResNet trained with standard cross-entropy loss and SGD. The model reached zero training error around epoch 43 and was then trained further into the terminal phase of training. I tracked paper-style metrics for NC1–NC4 using penultimate-layer features. The results show strong evidence for neural collapse, even in this simplified setting: within-class variation decreases substantially, classifier weights align strongly with class means, and disagreement with nearest-class-center prediction goes to zero. The simplex ETF trend is also present, though less complete than in the original large-scale study.

The reproduction supports the paper's central idea: training past zero error is not wasted computation. Even after classification is already correct, the feature space continues reorganizing into a more symmetric and interpretable structure. Personally, I like to see it as the model going from a good guesser, to a confident explainer, hence a better model.

1. Introduction

Deep classifiers are often treated as black boxes. They classify well, but the geometry of the learned representation is hard to see directly. The paper by Papyan, Han, and Donoho asks a sharp question: *what happens to the last-layer representation after training error has already reached zero?*

While initially surprising due to the black-box nature of deep learning, the result becomes more intuitive when analyzed through feature geometry. During the *terminal phase of training* (TPT), where training error is already zero but cross-entropy loss continues decreasing, the feature space does not stop changing. Instead, it converges toward a highly structured geometry called *neural collapse*. The paper reports this across seven benchmark datasets and three major architectures.

This project reproduces that qualitative phenomenon on a smaller scale. Using MNIST and a Mini-ResNet rather than ImageNet-scale experiments. The goal is not to match the original paper numerically. The goal is to test whether the same geometric trends appear in a simple, controlled setting using the same core ingredients: image classification, cross-entropy training, training beyond zero error, and analysis of penultimate-layer features.

2. Summary of the Paper

The paper studies a standard deep classifier of the form

$$\hat{y}(x) = \arg\max_{c'} \left(w_{c'}^\top h(x) + b_{c'} \right),$$

where $h(x) \in \mathbb{R}^p$ is the learned feature vector, and (W, b) is the final linear classifier.

The key claim is that during TPT, four connected properties emerge:

- **NC1: Variability collapse:** features within each class collapse toward their class mean.
- **NC2: Simplex ETF structure:** centered class means become equal in norm and equally separated in angle.
- **NC3: Self-duality:** classifier weights align with the centered class means.
- **NC4: Nearest class center behavior:** the classifier becomes equivalent to nearest-class-center (NCC) prediction.

The paper shows this behavior on MNIST, FashionMNIST, SVHN, CIFAR10, CIFAR100, STL10, and ImageNet using VGG, ResNet, and DenseNet models. It also reports that test accuracy and robustness can continue improving during TPT.

The central takeaway is simple: once a classifier has already fit the data, continued training still matters because it simplifies the representation geometry.

3. Core Ideas and Equations

Let $h_{i,c}$ denote the penultimate-layer feature for the ith training example in class c. The class mean and global mean are

$$\mu_c = \frac{1}{N} \sum_{i=1}^{N} h_{i,c}, \qquad \mu_G = \frac{1}{CN} \sum_{c=1}^{C} \sum_{i=1}^{N} h_{i,c}.$$

The paper measures geometry through the within-class and between-class covariance matrices:

$$\Sigma_W = \mathrm{Ave}_{i,c}\big[(h_{i,c} - \mu_c)(h_{i,c} - \mu_c)^\top\big],$$

$$\Sigma_B = \mathrm{Ave}_c\big[(\mu_c - \mu_G)(\mu_c - \mu_G)^\top\big].$$

3.1 NC1: Within-Class Collapse

A standard metric is

$$\mathrm{Tr}(\Sigma_W \Sigma_B^\dagger).$$

As this decreases, within-class spread becomes small relative to between-class structure.

3.2 NC2: Simplex ETF Structure

After centering by μ_G, the class means approach a simplex ETF. This means two things:

- their norms become equal,
- pairwise cosines approach $-1/(C-1)$.

For MNIST, $C = 10$, so the target cosine is $-1/9$.

3.3 NC3: Self-Duality

If

$$\dot{M} = [\mu_1 - \mu_G, \ldots, \mu_C - \mu_G],$$

then NC3 is measured by

$$\left\| \frac{W^\top}{\|W\|_F} - \frac{\dot{M}}{\|\dot{M}\|_F} \right\|_F .$$

When this becomes small, the classifier weights and centered class means have nearly the same geometry.

3.4 NC4: Nearest-Class-Center Behavior

NC4 says the network prediction becomes equivalent to

$$\arg\min_{c'} \|h - \mu_{c'}\|_2^2.$$

So although the model is trained as a deep network with a linear head, its final behavior becomes nearest-class-center classification.

Together, NC1–NC4 say that the final-layer geometry becomes much simpler than the network itself.

4. Reproduction Experiment

4.1 Experimental Goal

The goal was to reproduce the paper's main qualitative phenomenon under limited compute. That means preserving the essential setup while reducing scale. The experiment (code provided) can be run on the browser using commodity hardware in Google Colab.

4.2 Dataset and Preprocessing

I used the full MNIST training set (60,000 images) and test set (10,000 images). Images were padded from 28×28 to 32×32, converted to tensors, and normalized with the standard MNIST mean and standard deviation.

4.3 Model

I used a Mini-ResNet instead of a full ResNet18. The model contains:

- an initial convolutional stem,
- three residual stages,
- adaptive average pooling,
- a penultimate feature layer of dimension 32,
- a final linear classifier with 10 outputs.

The model has 176,730 parameters.

This design keeps the residual-network character while making feature analysis much cheaper.

4.4 Training

The network was trained for 80 epochs using cross-entropy loss and SGD with momentum, weight decay, and a multi-step learning-rate schedule. Training accuracy first reached 100% at approximately epoch 43. I treated this as the start of TPT and continued training through epoch 80.

4.5 Feature Analysis

At selected checkpoints (epochs 1,2,3,5,8,12,16,24,32,48,64,80), I collected penultimate-layer features, logits, and labels from the training set and computed:

- class means μ_c,
- global mean μ_G,

- within-class covariance Σ_W,
- between-class covariance Σ_B,
- paper-style metrics for NC1–NC4.

The tracked quantities were:

1. NC1 trace ratio: $\mathrm{Tr}(\Sigma_W \Sigma_B^\dagger)$,
2. NC2 norm coefficient of variation,
3. NC2 cosine ETF error,
4. NC3 weight–mean Frobenius distance,
5. NC4 mismatch between network predictions and NCC predictions.

5. Results

5.1 Training Dynamics

The experiment did enter the correct regime. Training accuracy reached 100% around epoch 43, while cross-entropy loss continued decreasing afterward. This is exactly the terminal phase of training studied in the paper.

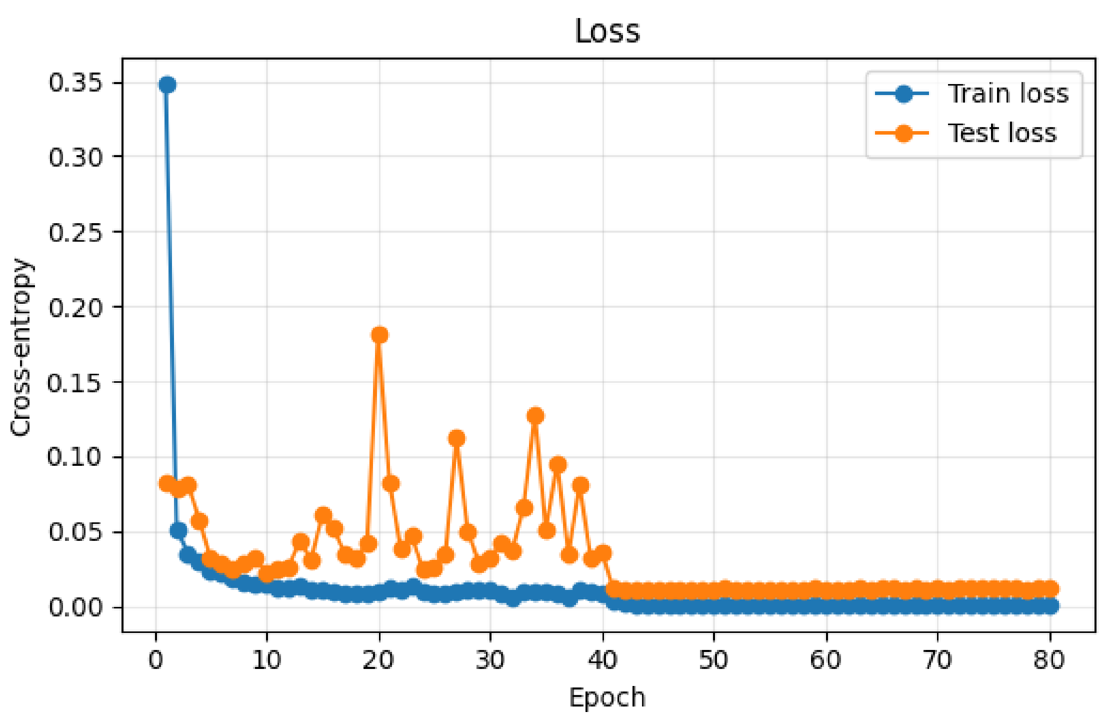

Figure 1: **Training dynamics on MNIST.** Training error reaches zero around epoch 43, marking the start of terminal phase of training. After that point, cross-entropy loss continues decreasing even though the model already classifies all training examples correctly.

5.2 Neural Collapse Metrics

The geometric metrics moved in the direction predicted by the paper.

NC1. The trace ratio decreased from about 2.679 at the first checkpoint to 0.307 at the last checkpoint. This shows substantial reduction in within-class variability relative to between-class structure.

NC2. The cosine ETF error decreased from about 0.320 to 0.167, and the norm coefficient of variation decreased from about 0.216 to 0.061. This shows that class means became more equal in norm and more regular in angle, although not perfectly ETF-like.

NC3. The weight–mean distance decreased from about 0.197 to 0.016. This is strong evidence that the classifier weights aligned closely with the centered class means.

NC4. The mismatch between network predictions and NCC predictions decreased from about 2.13% to 0%. By the end of training, the learned classifier and NCC rule agreed on all analyzed examples.

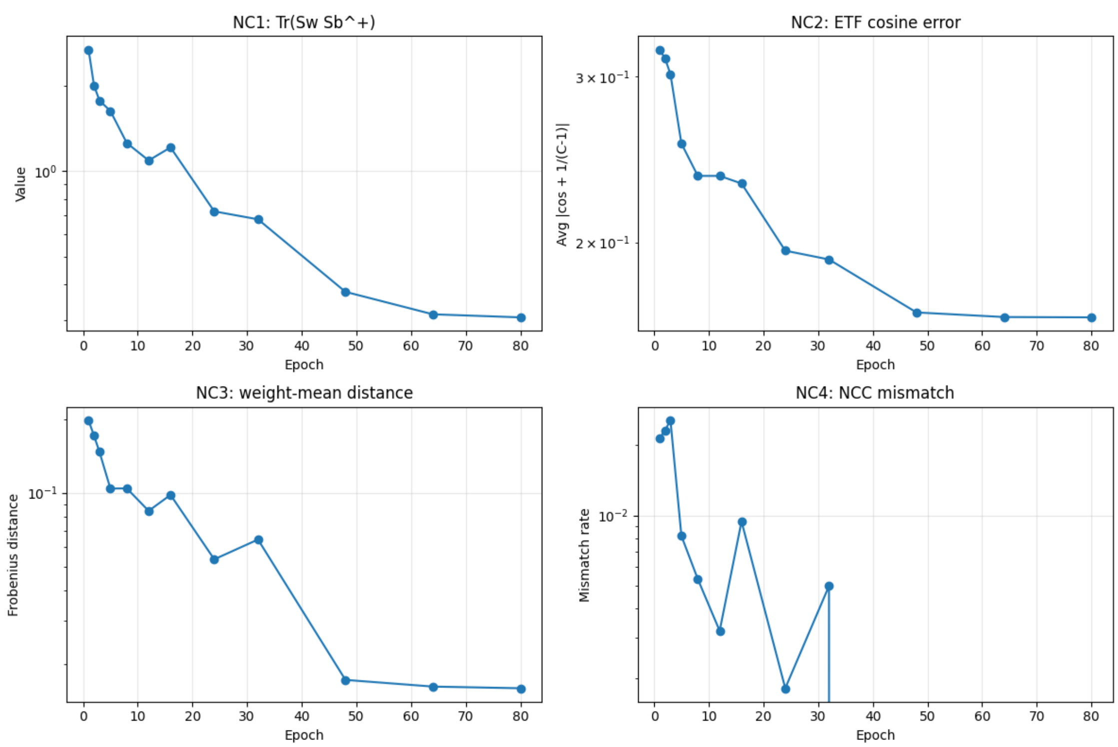

Figure 2: **Evolution of neural collapse metrics.** During and after TPT, NC1 decreases, NC3 decreases strongly, and NC4 goes to zero. NC2 also improves, showing movement toward simplex ETF geometry, though convergence is incomplete in this smaller setting.

Table 1: **First and last checkpoint values of the tracked metrics.**

Metric	**First checkpoint**	**Last checkpoint**
NC1: $\mathrm{Tr}(\Sigma_W \Sigma_B^\dagger)$	2.679	0.307
NC2: cosine ETF error	0.320	0.167
NC2: norm CoV	0.216	0.061
NC3: weight–mean distance	0.197	0.016
NC4: NCC mismatch	0.0213	0.0000

Overall, the reproduction captures the main qualitative pattern of the paper. The strongest evidence is in NC3 and NC4, while NC2 is present but incomplete.

6. Critical Evaluation

This reproduction supports the paper's central claim, but it is important to state clearly what was and was not reproduced.

6.1 What Worked Well

The simplified experiment successfully reproduced the main phenomenon:

- training continued beyond zero error,
- within-class variability decreased,
- classifier weights aligned with class means,
- classifier behavior converged to NCC.

These are not vague trends. NC3 became very small, and NC4 reached zero mismatch. That is strong qualitative evidence for neural collapse.

6.2 Main Discrepancy

Our replication shows all the trends how we would expect them. however there is a slight discrepancy in the experiments NC2. The class means clearly moved toward simplex ETF structure, but the convergence was not as complete. This is not surprising, Neural collapse is an asymptotic phenomenon; metrics approach their limits but do not reach them exactly in finite training. Notably, however, the original paper reaches a much smaller cosine error (0.01-0.001) for NC-2, than us, making a stronger argument for a simplex ETF structure.Given used larger models, more datasets, and significantly longer or more optimized training. My setting used a compact network, one dataset, and only 80 epochs.

So the fair conclusion is not that the paper's claim does not replicate. The fair conclusion is that the smaller setup reproduces the direction of the effect, but not the full, blatant geometry.

6.3 Other Limitations

This project also has other limits:

- only one dataset was used,
- only one main model was used,
- the experiment was effectively a single-run reproduction,
- adversarial robustness from the original paper was not reproduced.

These limits matter, but they do not undermine the main qualitative result.

6.4 Interpretation

The most important point of neural collapse is that the inside of a model is not arbitrary. Throughout setups, SGD with cross-entropy empirically appears to prefer a highly structured representation. A deep nonlinear network ends up behaving like a much simpler geometric classifier.

That is why the paper matters. It gives a way to look inside a trained classifier and see order instead of only complexity.

7. Reflection

The main thing I learned from this project is the insight that can be derived by analyzing the vectors, especially of class, means in a network.I had earlier known that these vectors had deep meaning. For example in a networks that do word embeddings, the vector difference between "Queen" and "King" would be similar to that of "girl" and "boy". This paper made it clear to me that networks are turning classes into vectors in space, where each direction points to some deep embedded characteristic. I believe that neural collapse shows strong evidence of this, as the best models (trained deep into TPT) are the ones that form vectors that are maximally and equally apart angle-wise. Meaning they form the greatest distinction between classes both mathematically and hence Ideologically.

The hardest part to understand at first was why nearest-class-center behavior would emerge from a learned linear classifier. That became clearer once I understood NC2 and NC3 together: if the class means become symmetric and the weights align with them, then NCC behavior is a natural consequence.

I also learned that replicating a machine learning paper not as intimidating as it may seem. Once the setting, goals and claims are understood clearly. In this case, those objects were the general process of deep neural nets, class means, within-class covariance, and between-class covariance. Once those were clear, the rest of the paper became much easier to follow.

8. Conclusion

This report reproduced the central qualitative phenomenon of neural collapse on MNIST using a Mini-ResNet. After the model reached zero training error, the feature geometry continued evolving during terminal phase of training. Within-class variability decreased, class means became more regular, classifier weights aligned with class means, and classifier behavior converged to nearest-class-center prediction.

The reproduction does not match the scale of the original paper, and simplex ETF convergence was only partial. Still, the main conclusion survives clearly: training beyond zero error is not merely reducing loss numerically. It is reorganizing the learned representation into a simpler and more symmetric structure.

Code and Presentation Resources

For transparency and reproducibility, all implementation code and the presentation delivered on April 9, 2026 in the University of Alberta ECE 447 course (instructor: Bahareh Tolooshams) are publicly available:

https://github.com/aliiqbal24/Neural-Collapse

References

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